

This dissertation examines how artificial intelligence interacts with human beatbox and, more broadly, with contemporary vocal and musical practices. Beatbox is taken here not as a series of mouth noises, but as an art form where a single body uses breath, gesture and voice to build rhythm and texture in real time. At the same time, AI tools now make it possible to imitate voices, generate patterns and produce highly polished sound in a few clicks. This raises a central question: when these technologies enter the field of vocal arts, do they replace the performer by imposing new technical standards, or can they extend beatbox and other sound practices while keeping human presence clearly visible?

The first part returns to the voice and to beatbox as a full art form, placing it alongside other musical and sound design practices (singing, electronic music, studio work). It combines basic elements of vocal physiology with approaches to listening and attention in order to describe what is specific about a live beatbox or vocal performance. The second part analyses what AI systems change in practice: illusions in listening, the search for clean and standardised sound, the comfort of instant answers, the role of platforms and formats, and the way authorship can become blurred. The third part proposes conditions for a demanding alliance between AI and human performers. It outlines scenarios in which AI remains an instrument or a partner rather than a stand-in, formulates minimal criteria of effort, decision and traceability that should remain perceivable, and draws practical recommendations from case studies, competitions and an interview with a professional musician. Overall, the dissertation argues that AI can widen the field of vocal art only if we can still answer clearly what is heard, who is acting, and where the human body remains present in the performance.